

Research Proposal

Risk-Asymmetric Selective Prediction for Commercial Aircraft Engine Remaining Useful Life (RUL) Estimation

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Background and Introduction to the Area

Prognostics and Health Management (PHM) has emerged as a fundamental pillar of modern industrial asset management, especially in safety-critical domains such as aerospace engineering. The primary objective of asset prognostics is to accurately estimate the Remaining Useful Life (RUL). The RUL is defined as the time or number of flight cycles left before the equipment component can no longer operate at or above a safe operational threshold.

Historically, maintenance in aircraft engines has relied on hard, schedule-based interventions (e.g., servicing an engine every 500 cycles) or reactive maintenance (fixing a component after failure). Schedule-based frameworks do not permit catastrophic events to occur, but they are economically inefficient because parts with useful service life are often discarded.

With the advent of high-frequency Internet of Things (IoT) sensor arrays on modern turbofan engines, the industry has quickly shifted to Predictive Maintenance (PdM). Modern machine learning (ML) architectures, including Deep Convolutional Networks, Long Short-Term Memory (LSTM) networks and Transformers are trained on

multivariate time-series sensor streams (e.g., temperature, pressure, fan speeds) to directly map degradation profiles to a deterministic RUL point estimate.

But the main problem in deploying such models in live aviation operations is this: typical deep learning systems rely on a closed-world assumption and produce single-point predictions without any measure of confidence. Unfortunately, point predictions without formal Uncertainty Quantification (UQ) are operationally non-viable in an environment where over-optimistic prediction can result in inflight engine failure and under-optimistic prediction results in millions of dollars of premature maintenance costs.

The paper treats the evolution of RUL estimation from simple regression to a reliable self-aware Selective Prediction Framework. The system is able to identify its own operational boundaries by providing distribution-free uncertainty bounds for deep learning architectures and automatically ceding highly uncertain predictions to human experts.

Literature Review

Data-driven RUL estimation with the benchmark NASA C-MAPSS dataset has been widely studied in the academic literature from statistical modeling to very complex deep learning. Recently, a distinct shift towards quantification of the reliability of these models has emerged. Five key papers outline the development and existing gaps within this domain:

1. Point Estimation Foundations

Babu, G. S., Zhao, P., & Li, X.-L. (2016). "Deep Convolutional Neural Network Based Regression Approach for Remaining Useful Life Estimation." International Conference on Neural Information Processing.

Contribution: This seminal paper was one of the first to move away from hand-crafted statistical features by directly applying Deep Convolutional Neural Networks (CNN) on multivariate time-series data from the C-MAPSS dataset. The authors demonstrated that an auto sliding-window feature extractor can achieve better performance than traditional Support Vector Regression (SVR) and Multi-Layer Perceptron (MLPs).

Limitation: Deterministic architecture only. Predictions are treated the same, with no way to indicate that test data is outside the training data distribution.

2. Bayesian Time-Series Uncertainty

Chen, S., He, J., Wen, P., et al. (2023). Convolutional Bayesian Long Short-Term Memory Neural Network for Remaining Useful Life Prognostics and Uncertainty Quantification of Aircraft Engines IEEE Conference on Prognostics & Health Management.

Contribution: The authors have developed a hybrid Convolutional Bayesian LSTM (CB-LSTM) networks. The model captures Epistemic Uncertainty by placing a prior distribution over the neural network weights and doing variational inference. (model uncertainty caused by limited training data) but also providing accurate point predictions.

Limitation: Bayesian Neural Networks (BNNs) require computing a complex posterior distribution that is computationally expensive and slow for real-time edge streaming. In addition, the resulting prediction intervals are very sensitive to the choice of an appropriate prior distribution and to model misspecification.

3. Deep Ensembles and the negative-loglikelihood

Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017). "Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles." NeurIPS. (Applied to C-MAPSS via OpenReview, 2024)

Contribution: This framework overcame the computational challenges of BNNs by training an ensemble of independent deterministic LSTMs. Importantly, we modified the models to output a mean (μ) and variance (σ^2) for each step and trained them to minimize a Negative Log-Likelihood (NLL) loss function to learn Aleatoric Uncertainty. (inherent data noise)

Limitation: Deep ensembles are very scalable, but do not have any theoretical guarantees. However, if the entire ensemble is uniformly overconfident on a highly anomalous out-of-distribution engine profile, the aggregated uncertainty metric can still fail to flag the danger.

4. Distribution-Free Statistical Guarantees

Javanmardi, H., & Hüllermeier, E. (2023). "Conformal Prediction Intervals for Remaining Useful Lifetime Estimation." International Journal of Prognostics and Health Management.

Contribution: This paper proposed the Conformal Prediction (CP) framework for the RUL problem. CP is a post-processing wrapper that can be applied to any pre-trained model (e.g. Gradient Boosting, CNNs). With the mild assumption of data exchangeability, it provides valid prediction intervals that ensure that the true RUL will lie in the bounded range at a user-specified confidence level (e.g., 90%).

Limitation: The generated intervals are symmetric about the prediction point, so that early maintenance penalties and late maintenance catastrophes are given equal mathematical weight.

5. Risk-Aware Conformalization

Robinson, C. (2026). "Remaining Useful Life Estimation for Aircraft Engines with Risk-Aware Prediction Intervals via Conformalized Quantile Regression." International Journal of Prognostics and Health Management.

Contribution: The symmetry problem has been addressed in a recently published paper with an asymmetric formulation of Conformalized Quantile Regression (CQR). It intentionally skews the miscoverage error bound towards safety, in an effort to have the model minimize the risks of late maintenance interventions.

Limitation: The intervals are risk-aware, but the model does not have an automated Selective Prediction mechanism. It does not actively address the operational problem of dynamically routing high-risk/high-uncertainty engine assets away from automation to human-in-the-loop review systems.

Problem Statement

Despite the high accuracies of modern deep learning models on RUL benchmark datasets, existing methodologies fail to properly address the prediction uncertainty for safe production deployment. This is due to three critical gaps

Lack of Strict Operational Selective Prediction Architecture

Current research treats Uncertainty Quantification only as a metric to evaluate and plot. In the context of real aviation infrastructure, a model needs to know when to refrain. There is a striking lack of an algorithmic framework for converting a broad confidence interval or high epistemic variance into an automatic "abstain-and-escalate" decision for safely handing over the asset to human engineers.

Symmetric Risk Vulnerability

Standard probabilistic neural networks use loss functions like Mean Squared Error (MSE) or symmetric Negative Log-Likelihood (NLL) that penalize over-prediction and under-prediction equally. In fact, an RUL of 30 cycles is predicted when the actual RUL is 10, leading to catastrophic inflight component failure. On the other hand, predicting an RUL of 10, when the true RUL is 30 will only result in a premature, safe maintenance check. Current uncertainty frameworks do not map their confidence intervals to these asymmetric operational risk realities.

Vulnerability to Non-Stationary Sensor Degradation

Real-world deployments introduce environmental noise, sensor degradation, and communication drops. Current models assume clean test data. When Deterministic models, when faced with out-of-distribution (OOD) sensor drifts, produce highly confident but completely wrong predictions, and current Bayesian models are not up to the task because of their extensive computational overhead.

Proposed Method

To mitigate these issues, this paper proposes a **Risk-Asymmetry Selective Prediction Framework** built on top of a high-performance temporal backbone validated with a post-hoc statistical calibration layer.

Instead of predicting a single number for an engine's Remaining Useful Life (RUL), the system generates a safety-optimized confidence window and uses a decision engine to direct high-risk scenarios to human operators.

The Selective Forecasting Pipeline: Bridging AI Prediction and Operational Risk

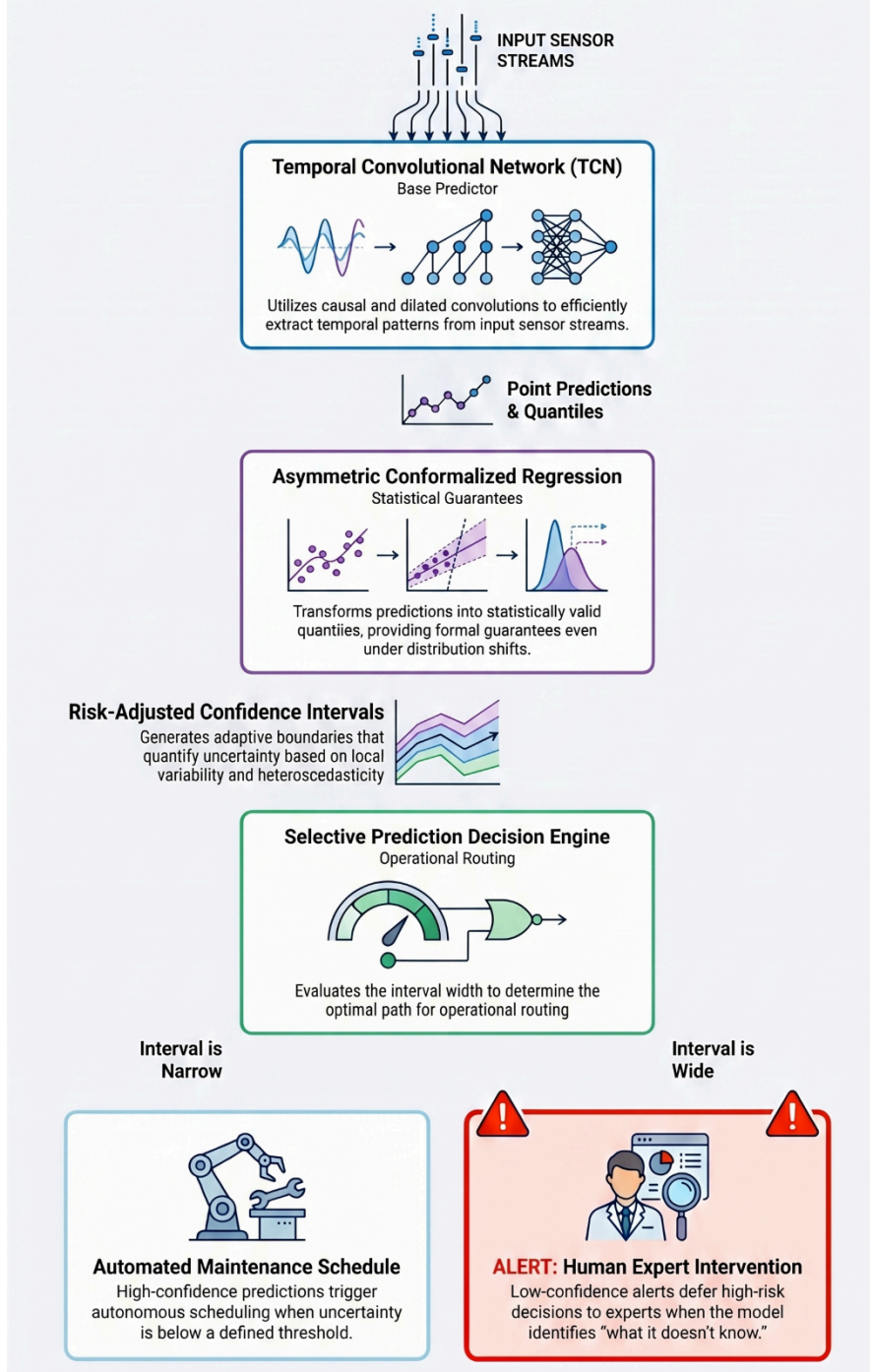


Figure 01

Step 1: The Base Predictive Network

The underlying predictive engine will be a **Temporal Convolutional Network (TCN)** or a Gated Recurrent Unit (GRU) network trained with Quantile Regression. Instead of optimizing for a single mean RUL prediction, the model will simultaneously predict three target boundary predictions, namely a low-percentile boundary, a median value, and a high-percentile boundary. This gives us an initial uncalibrated range of where the true remaining life of the engine is likely to be based on historical patterns.

Step 2: Risk-Asymmetric Conformal Calibration

We will also apply **Conformalized Quantile Regression (CQR)** on a separate calibration subset of the data to obtain strict distribution-free guarantees. This adds a postprocessing layer that quantifies the model's output boundaries based on its true performance on held-out data.

Crucially, we propose an asymmetric penalty scheme here. In aviation, overestimating an engine's life is very dangerous; underestimating it is an early, safe inspection. The calibration layer is weighted to strongly penalize over-optimistic errors vs under-optimistic ones. This means that the conformal confidence intervals should defensively widen when the network sees ambiguous signatures of degradation or unfamiliar sensor noise.

Step 3: The Selective Prediction Routing Mechanism

The last layer of the proposed architecture is a decision boundary that is defined in a programmatic manner based on the width of the calibrated prediction interval. If the model is very confident, the upper and lower bounds will be close together. If the model is confused or has an anomaly, the gap will be big.

The system compares this interval width with a pre-set risk threshold.

- **Narrow interval (Low Uncertainty)** : The automated system takes the median prediction and updates the aircraft digital twin maintenance database so that servicing is scheduled normally.
- **If the interval is large (High Uncertainty)**, the system does not make any autonomous decision and marks the asset as a "High Uncertainty Asset" and raises an immediate alert to send the case to a human maintenance engineer for manual inspection review.

List of Available Datasets

The project will be trained, calibrated, and stress-tested using the gold-standard simulation datasets provided by the NASA Prognostics Center of Excellence:

1. **NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation)**
 - **Structure:** It is composed of four sub-datasets (FD001, FD002, FD003, FD004) simulating run-to-failure trajectories of turbofan engines.
 - **Variables:** 21 sensor channels (including total temperature, pressure, bypass ratio, and bleeds) and 3 operational settings.
 - **Usefulness for this project:** We will take FD001 (single operating condition, single failure mode) as our core baseline pipeline. We will then explicitly evaluate our uncertainty framework on FD004 (six operating conditions, two complex failure modes), as the multi-modal operating environments naturally induce high epistemic uncertainty.
2. **NASA N-MAPSS (New Commercial Modular Aero-Propulsion System Simulation)**
 - **Structure:** Modernized version of the classic dataset, with much larger data scales with real-world flight profile traces (climb, cruise, descent) instead of simplified cyclic repetitions.
 - **Variables:** Thermodynamic parameters, sub-component degradation profiles at a high level of detail
 - **Usefulness of N-MAPSS for this project:** N-MAPSS will be our main platform for testing Out-of-Distribution (OOD) robustness. We are able to train the model with normal flight profiles and explicitly test the selective prediction layer on anomalous, high-stress flight profiles to see if the model abstains in a safe manner when the accuracy drops.